Department of Computer Science and Engineering
National Institute of Technology Karnataka, Surathkal

2421S015

Mid-Semester Examination (February 2025)

Course Name: Quantum Cryptography
Semester: II
Duration: 1 Hour 30 Minutes

Course Code: CS826
Course: M.Tech. (CSE-IS)
Marks: 40

Note: Answer all the questions.

Q1. (a) Consider the complex number z=1+2i. **[08]**

- **(i)** Find the real component *z* of $\Re(z)$.
- **(ii)** Find the imaginary component z of $\Im(z)$.
- **(iii)** Write z in the polar form $re^{i\theta}$.
- **(iv)** Find the conjugate z*.

(b) A qubit is in the state **[04]**

$$\frac{1+i\sqrt{3}}{3}|0\rangle + \frac{2-i}{3}|1\rangle.$$

If you measure the qubit, what is the probability of getting

(a) $|0\rangle$?
(b) $|1\rangle$?

Q2. (a) A qubit is in the state

$$A\left(2e^{i\pi/6}|0\rangle - 3|1\rangle\right).$$

(i) Normalized the state (i.e., find A.). **[04]**

(ii) If you measure the qubit, what is the probability that you get $|0\rangle$? **[04]**

Q3. Consider a map *U* that transforms the Z-basis states as follows.

$$U|0\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{\sqrt{3}+i}{4}|1\rangle,$$

$$U|1\rangle = \frac{\sqrt{3}+i}{4}|0\rangle - \frac{\sqrt{3}+3i}{4}|1\rangle.$$

Say $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is a normalized quantum state, i.e., $|\alpha|^2 + |\beta|^2 = 1$.

(i) Calculate $U|\psi\rangle$. **[05]**

(ii) From your answer to (i), is *U* a valid gate? Explain your reasoning. **[05]**

Q4. (a) The Hadamard gate turns $|0\rangle$ into $|+\rangle$, and $|1\rangle$ into $|-\rangle$.

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle,$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle.$$

From the definition of Hadamard gate, prove that H|+> = |0>, H|-> = [redacted]|1> and H|i> = [redacted]|-i>. **[02+02+02]**

(b) Consider each logical classic gate with inputs A and B, outputs C and D, and the truth table below. Is each gate a valid quantum gate? Why? **[02+02]**

(a)

A	B
0	0
1	1

(b) CD

A	B
0	1
1	1

C D

C

D

H|i> = |-i>

|00> + |11>

|01> + |11|

Department of Computer Science and Engineering
National Institute of Technology Karnataka, Surathkal

End-Semester Examination (May 2025)

Course Name: Quantum Cryptography

Course Code: CS826

Semester: II

Course: M.Tech. (CSE-IS)

Duration: 03 Hours

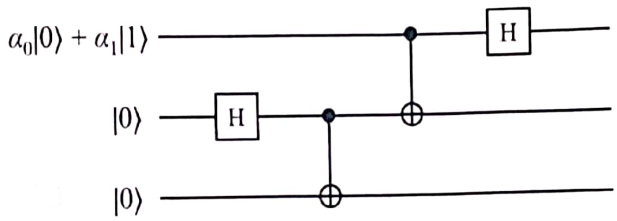
Marks: 80

Note: Answer all the questions.

Q1. **(a)** Explain Shor's Algorithm and describe through a suitable example how it can pose a threat to RSA. **[6]**

(b) A SWAP gate takes two inputs x_1 and x_2 and outputs x_2 and x_1; i.e., it swaps the values of two registers. Show how to build a SWAP gate using only CNOT gates. **[6]**

Q2. Consider the following quantum circuit:



An equivalent description of the circuit (calling the registers x_1, x_2, x_3) is:

1. Initialize x_1 to $\alpha_0\,|0\rangle + \alpha_1\,|1\rangle$
2. Initialize x_2 to $|0\rangle$
3. Initialize x_3 to $|0\rangle$
4. Hadamard(x_2)
5. CNOT(x_2, x_3)
6. CNOT(x_1, x_2)
7. Hadamard(x_1)

a) Determine with proof the state of the three qubits at the end of the circuit's operation. **[05]**

b) If we then measure the three qubits, give the outcomes and their probabilities that arise. **[05]**

Q3. Calculate the following inner products. **[03+03+03]**

a) $\langle 10|11\rangle$.

b) $\langle +-|01\rangle$.

c) $\langle 1+0|1-0\rangle$.

Q4. (a) If you measure the left qubit, what outcomes can you get, what are the corresponding probabilities of those outcomes, and what does the state collapse to for each outcome? Is this state a product state, partially entangled state, or maximally entangled state? **[03+03+03]**

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle).$$

(b) Alice and Bob share an EPR-pair,

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Suppose they each measure their qubit with an X-observable (which corresponds to a particular projective measurement with possible outcomes +1, −1). Find the following. **[10]**

(a) P_{00}.

(b) P_{01}.

(c) P_{10}.

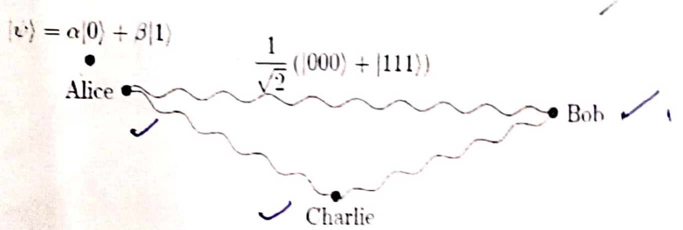
(d) P_{11}.

(e) $E(A',B') = P00 - P01 - P10 + P11$.

Q5. Alice wants to teleport a qubit in an unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to Charlie, and Bob is helping her. They share three entangled qubits in the GHZ state:

$$|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle).$$

The left qubit is Alice's, the middle qubit is Bob's, and the right qubit is Charlie's.



Altogether, the initial state of the system is

$$|\psi\rangle|\mathrm{GHZ}\rangle = (\alpha|0\rangle + \beta|1\rangle)\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$
$$= \frac{1}{\sqrt{2}}(\alpha|0000\rangle + \alpha|0111\rangle + \beta|1000\rangle + \beta|1111\rangle)$$

So, the left two qubits are Alice's, the second-to-right qubit is Bob's, and the right qubit is Charlie's.

(a) Show that if Alice applies CNOT to her qubits (so the far left qubit is the control and the second-to-left qubit is the target) and then the Hadamard gate to her left qubit, the state of the system becomes **[5]**

$$\frac{1}{2}\left[|00\rangle(\alpha|00\rangle+\beta|11\rangle)+|01\rangle(\beta|00\rangle+\alpha|11\rangle)\right.$$
$$\left.+|10\rangle(\alpha|00\rangle-\beta|11\rangle)+|11\rangle(-\beta|00\rangle+\alpha|11\rangle)\right].$$

(b) Next, Alice measures her two qubits and makes the results known. What values can she get, with what probabilities, and what does the state collapse to in each case? **[5]**

Q6. (a) Explain the Crystal-Kyber public encryption algorithm. **[6]**

(b) Show that the following gates are reversible or irreversible. **[4]**

(i) OR. (ii) XOR. (iii) NAND. (iv) NOR

Q7. (a) Let's assume:

Alice's key:

A = 1 0 1 1 0 1 1 0

Bob's key:

B = 1 0 1 1 1 1 1 0

The above Alice and Bob raw keys contain discrepancies. Demonstrate how the Cascade protocol can be used to detect and correct errors. **[5]**

(b) Explain the BB84 Quantum Key Distribution protocol with the help of a suitable example. **[5]**

252PS032

Department of Computer Science and Engineering
National Institute of Technology Karnataka, Surathkal

Mid-Semester Examination (February 2026)

Course Name: Quantum Cryptography | **Course Code:** CS826
Semester: II | **Course:** M.Tech. (CSE-IS)
Duration: 1 Hour 30 Minutes | **Marks:** 25
Note: Answer all the questions.

Q1. (a) A qubit is in the following state: [2]

$$\begin{pmatrix} \sqrt{3}/2 \\ 1/2 \end{pmatrix}$$

If you measure this qubit in the Z-basis $\{|0\rangle, |1\rangle\}$, what states can you get and with what probabilities?

(b) Consider

$$|a\rangle = \frac{3+i\sqrt{3}}{4}|0\rangle + \frac{1}{2}|1\rangle.$$
$$|b\rangle = \frac{1}{4}|0\rangle + \frac{\sqrt{15}}{4}|1\rangle.$$

(i) Find $\langle a|b\rangle$. [1]
(ii) Find $\langle b|a\rangle$. [1]
(iii) What is the relationship between your answers to parts (a) and (b)? [1]

(c) Consider a qubit in the following state [4]

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle.$$

Consider measuring this qubit in the Y-basis $\{|i\rangle, |-i\rangle\}$ and the orthonormal basis $\{|a\rangle, |b\rangle\}$, where

$$|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle.$$
$$|b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle.$$

(i) Calculate $\langle i|\psi\rangle$.
(ii) Calculate $\langle -i|\psi\rangle$.
(iii) If you measure the qubit in the Y-basis, what states can you get and with what probabilities?
(iv) Calculate $\langle a|\psi\rangle$.
(v) Calculate $\langle b|\psi\rangle$.
(vi) If you measure the qubit in the $\{|a\rangle, |b\rangle\}$ basis, what states can you get and with what probabilities?

Q2. (a) The following two states are opposite points on the Bloch sphere:

$$|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle.$$
$$|b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle.$$

So, we can measure relative to them. Now consider a qubit in the state

$$|\varphi\rangle = \frac{1}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle.$$

(i) Write the qubit's state in terms of $|a\rangle$ and $|b\rangle$. [2]

(ii) If you measure the qubit in the basis $\{|a\rangle, |b\rangle\}$, what states can you get and with what probabilities? [2]

(b) Consider the following two states

$$|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle,$$
$$|b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle.$$

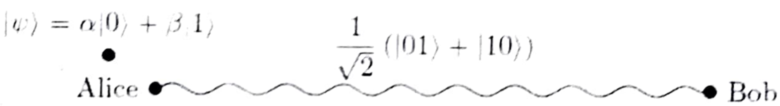
Prove these are opposite points of the Bloch sphere by finding their points in spherical coordinates (θ_a,φ_a) and (θ_b,φ_b). Verify that $\theta_b = \pi - \theta_a$ and $\varphi_b = \varphi_a + \pi$, which means they lie on opposite points of the Bloch sphere. [2]

Q3. Consider the following state of two qubits:

$$\frac{\sqrt{3}}{2\sqrt{2}}|00\rangle + \frac{1}{2\sqrt{2}}|01\rangle + \frac{1}{2\sqrt{2}}|10\rangle + \frac{\sqrt{3}}{2\sqrt{2}}|11\rangle$$

If you measure the left qubit, what outcomes can you get, what are the corresponding probabilities of those outcomes, and what does the state collapse to for each outcome? Is this state a product state, partially entangled state, or maximally entangled state? [1+1+1+1]

Q4 Alice wants to teleport a qubit in an unknown state $|\psi\rangle = \alpha|0\rangle+\beta|1\rangle$ to Bob. Instead of sharing two entangled qubits in the $|\Phi+\rangle$ state, they share two entangled qubits in the $|\Psi+\rangle$ state:



Altogether, the initial state of the system is

$$|\psi\rangle|\Psi^{+}\rangle = (\alpha|0\rangle + \beta|1\rangle)\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$= \frac{1}{\sqrt{2}}(\alpha|001\rangle + \alpha|010\rangle + \beta|101\rangle + \beta|110\rangle)$$

So, the left two qubits are Alice's, and the right qubit is Bob's.

(i) Show that if Alice applies CNOT to her two qubits, followed by H to her left qubit, the state of the system becomes [2]

$$\frac{1}{2}\big[|00\rangle(\beta|0\rangle + \alpha|1\rangle) + |01\rangle(\alpha|0\rangle + \beta|1\rangle)$$
$$+ |10\rangle(-\beta|0\rangle + \alpha|1\rangle) + |11\rangle(\alpha|0\rangle - \beta|1\rangle)\big].$$

(ii) Next, Alice measures both of her qubits. What values can she get, with what probabilities, and what does the state collapse to in each case? [2]

(ii) Finally, Alice tells Bob the results of her measurement. For each possible result, what should Bob do to his qubit so that it is $\alpha|0\rangle+\beta|1\rangle$, the state that Alice wanted to teleport to him? [2]

Department of Computer Science and Engineering
National Institute of Technology Karnataka, Surathkal

Class Test-3 (April 2026)

Course Name: Quantum Cryptography
Semester: II
Duration: 50 Minutes

Course Code: CS826
Course: M.Tech. (CSE-IS)
Marks: 20

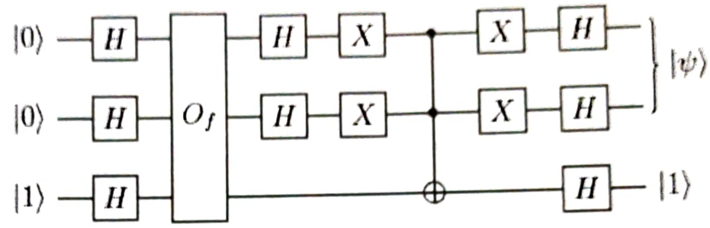
Note: Answer all the questions.

Q1. Consider a search space of dimension N = 4 with its elements encoded in binary {00,01,10,11}. Suppose you are searching for the element z = 11.

a) Construct the circuit implementing the quantum oracle $O_f : |x\rangle|y\rangle \to |x\rangle|y \oplus f(x)\rangle$ for the function: **[3]**

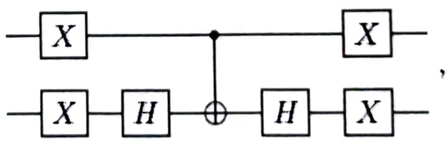
$$f(x) = \begin{cases} 1 & \text{for } x = z \\ 0 & \text{otherwise} \end{cases}$$

b) We can now construct the quantum circuit which performs the initial Hadamard transformations and a single Grover iteration G:



a) Compute the output state. **[3]**
b) What happens after we measure the output in the computational basis? **[1]**
c) How many times do we have to repeat G to obtain z in this example? **[1]**

Q2. The target (lower) qubit is a catalytic qubit used to implement a phase-kickback unitary on the address (top) qubits. Show that the circuit below implements the conditional phase shift transformation $2|00\rangle\langle 00| - I$, up to a global phase. **[3]**



b) Suppose we run Simon's algorithm on the following function $f(x) : \{0,1\}^3 \to \{0,1\}^3$.

$$f(000) = f(111) = 000$$
$$f(001) = f(110) = 001$$
$$f(010) = f(101) = 010$$
$$f(011) = f(100) = 011$$

Where $f(x)$ is $2-\text{to}-1$ and $f(x_i) = f(x_i \oplus 111)$ for all $i \in \{0,1\}^3$; therefore the period is $a = 111$.

a. What is the initial input of Simon's algorithm? $|000\rangle|000\rangle$ **[3]**

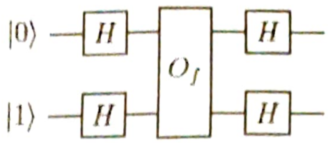
$\begin{pmatrix}1\\0\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}$ $\frac{1}{\sqrt2}(|0\rangle - |1\rangle)$

$\frac{1}{\sqrt2}\begin{pmatrix}1\\-1\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix} = \begin{pmatrix}-1\\1\end{pmatrix}$ $\frac{1}{\sqrt2}\begin{pmatrix}1\\0\end{pmatrix}-\begin{pmatrix}0\\1\end{pmatrix}$ 000 111 111

$\begin{pmatrix}0\\1\end{pmatrix}-\begin{pmatrix}1\\0\end{pmatrix}$ $|1\rangle - |0\rangle$ $\begin{pmatrix}1\\-1\end{pmatrix}$ $-|1\rangle|-\rangle$ $\frac{(|0\rangle + |1\rangle)}{\sqrt2}|-\rangle$

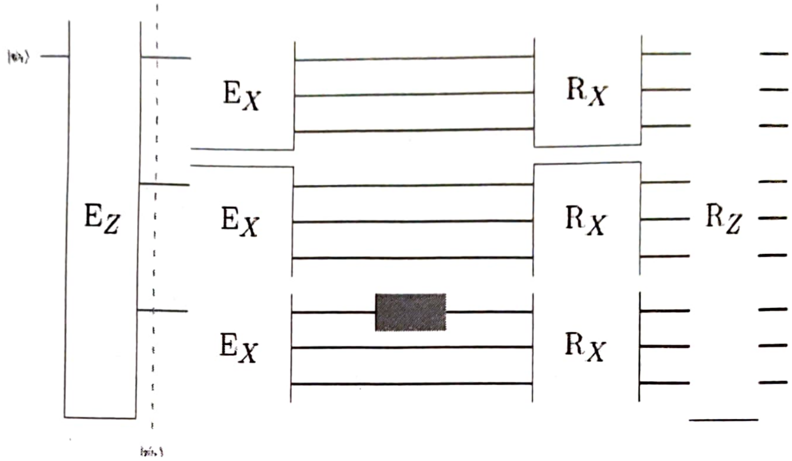
$-|-\rangle$

Q3. Consider the following circuit:



The oracle O_f is a two-qubit gate that maps $|x\rangle\,|y\rangle \rightarrow |x\rangle\,|y \oplus x\rangle$. By comparing to what we have seen in the Deutsch Algorithm, what is the classical function implemented by the oracle O_f, i.e. f(x)? Do you think f(x) will be balanced or constant? **[3]**

Q4. Shor's 9-qubit code allows us to encode our state in 9 qubits and determine whether any arbitrary single-qubit error has occurred, and where. Consider the following circuit:



where EZ and EX are the encoding circuits, and RX and RZ are the recovery circuits. The orange box in the circuit signifies a single qubit error occurring on the 7th qubit. Assuming that the initial state is $|\psi i\rangle = a\,|0\rangle + b\,|1\rangle$,

a. What is the state $|\psi_1\rangle$? **[3]**

Roll No.								

NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL

Department of Computer Science and Engineering

End Semester Examination

Program: M.Tech.(CSE-IS)
Course Code: CS826
Course Title: Quantum Cryptography
Date & Time: 30-04-2026 & 9 AM to 12 Noon
Duration: 03 Hours

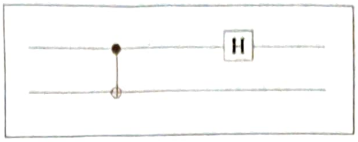
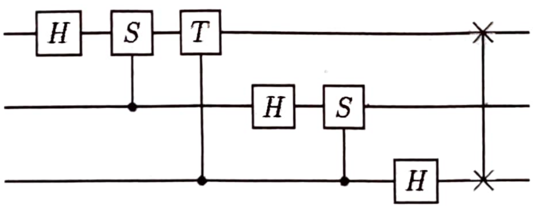
Year, Semester: 2026, Even
(L-T- P) C: (3-1-0)4

Max. Marks: 100
Weightage (%): 40

Note:

1) All the Questions are Compulsory.
2) Unless otherwise specified, assume suitable data.
3) Please add additional instructions as required.

Sl. No.	Question	Marks	CO
1	(a) Write a matrix $A \in C^{d\times d}$ with entries A_{ij} in the computational basis in bra-ket notation. (b) Use the resolution of the identity to determine the bra-ket form of an arbitrary matrix A. (c) What is the dimension of $C^d \otimes C^{d'}$? (d) Describe a quantum circuit with two gates that, starting from $\|00\rangle$, prepares the maximally entangled state $1/\sqrt{2}\,(\|00\rangle + \|11\rangle)$. Prove that your circuit acts as desired.	4×5 = 20	1
2	(a) Assume that both Alice and Bob know that the state that Alice wants to transmit is on the equator of the Bloch sphere, i.e. one of the states $\|\alpha\rangle = 1/\sqrt{2}\,(\|0\rangle + e^{-i\alpha}\|1\rangle)$ for $0 \le \alpha \le 2\pi$. Moreover, Alice knows the value α, i.e. she prepares the state herself. Let us define $\|\alpha^\perp\rangle = \|\alpha + \pi\rangle$. Show that $\|\alpha^\perp\rangle$ is orthogonal to $\|\alpha\rangle$ and express the computational basis states $\|0\rangle$ and $\|1\rangle$ in terms of $\|\alpha\rangle$ and $\|\alpha^\perp\rangle$.	10	2
	(b) Show that $\|\Phi^+\rangle = 1/\sqrt{2}\,(\|\theta\uparrow\rangle + \|\theta\downarrow\rangle)$, where $\|\theta\uparrow\rangle = \cos\theta\|0\rangle + \sin\theta\|1\rangle$ and $\|\theta\downarrow\rangle = -\sin\theta\|0\rangle + \cos\theta\|1\rangle$. i.e., verify that, $\|00\rangle + \|11\rangle = \|\theta\uparrow,\theta\uparrow\rangle + \|\theta\downarrow,\theta\downarrow\rangle$.	10	
3	(a) Consider the following 2-qubit state $$\|\Psi\rangle = \frac{1}{\sqrt{2}}\|00\rangle + \frac{1}{2}\|01\rangle + \frac{1}{2}\|11\rangle .$$ Suppose you measure the first qubit and obtain a measurement of $\|0\rangle$. (i) What is the state after this measurement? (ii) If you now measure the second qubit, what are the possible results and probabilities of those results? Suppose instead you measure the first qubit and obtain a measurement of $\|1\rangle$. (iii) What is the state after this measurement? (iv) If you now measure the second qubit, what are the possible results and probabilities of those results?	2.5×4 = 10	3
	(b) Alice wants to send two classical bits to Bob using one qubit (superdense coding), She needs only to transform her qubit into		

	the Bell State corresponding to the two bits she wants to send, then send her half to Bob (this requires a quantum channel). Bob can then recover Alice's two-bit message. Now suppose Alice want's to send Bob the message 01. Alice then applies Pauli matrix X to |φ+> to get |ψ+>: $$X\|\phi^+\rangle = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle) = \frac{1}{\sqrt{2}}(\|01\rangle + \|10\rangle) = \|\psi^+\rangle$$ Show how Bob can now recover Alice's message using the following Reverse Bell Circuit.  	10	
4	(a) Consider a 3-qubit system where the hidden string *s=101*. Find this string using the Bernstein–Vazirani algorithm.	10	
	(b) Using Shor's algorithm, find step-by-step the prime factors of the integer 15.	10	4
5	(a) You have eight bits, and using Deutsch's algorithm, you have found the parties of pairs of bits, shown below: $\underbrace{b_0, b_1}_{1},\ \underbrace{b_2, b_3}_{1},\ \underbrace{b_4, b_5}_{0},\ \underbrace{b_6, b_7}_{1}$ (i) What is the parity of all eight bits? (ii) How many queries to the oracle did it take to find the parity of all eight bits? (iii) In the worst case, how many queries does it take to solve the problem classically?	10	
	(b)Consider the following quantum Fourier circuit for three qubits  with S and T being the gates: $$S = \begin{pmatrix}1 & 0\\ 0 & e^{i\pi/2}\end{pmatrix} = \begin{pmatrix}1 & 0\\ 0 & i\end{pmatrix}, T = \begin{pmatrix}1 & 0\\ 0 & e^{i\pi/4}\end{pmatrix}$$ Suppose that we input the state $\|j\rangle = \|j_1 j_2 j_3\rangle$. What will be the output state?	10	5
